Playful Steps for Curious Minds!

Subject: AI & Digital Literacy	**Class:** Grade 2
Topic: Algorithms	**Duration:** 45 Minutes

■ Learning Objectives

- ■ Understand what an algorithm is using everyday examples.
- ■ Identify algorithms in daily activities such as morning routines and cooking.
- ■ Recognise how computers and robots use algorithms.
- ■ Create a simple algorithm of their own.
- ■ Appreciate that algorithms exist in nature as well.

1. What Is an Algorithm?

An **algorithm** is a set of steps to solve a problem or do a task. Think of it like a recipe for cooking your favourite dish, or the steps you follow to brush your teeth every morning. When you do things step by step in order, that is an algorithm! Algorithms help us do tasks the right way, every time.

■ **Did You Know?** When you follow your morning routine, you are already using an algorithm!

2. Why Are Algorithms Important?

Algorithms help us every day in many ways!

■ **Easier & Faster**	■ **Helps Computers & Robots**	■■ **Solves Problems**
Algorithms make tasks easier and faster — like getting ready for school quickly!	Computers and robots follow algorithms to know exactly what to do next.	Algorithms help us figure out solutions to problems we face every day!

3. Morning Routine Algorithm

Every morning, we follow the same steps in order. This is a perfect example of an algorithm in real life!

1	■■ Wake up
2	■ Brush your teeth
3	■ Get dressed
4	■ Eat breakfast

Following these steps every day is just like an algorithm — a set of steps done in order to complete a task!

4. Brushing Teeth — Step-by-Step

Follow these simple steps every morning and night for a bright, healthy smile! Our brushing routine is an algorithm — the same steps every time.

1	■ Wet your toothbrush
2	■ Put toothpaste on
3	■ Brush all your teeth
4	■ Rinse and smile!

5. Making a Sandwich — Algorithm in Cooking ■

Follow these yummy steps to make a delicious sandwich. Every time you follow the same steps to make food, you are using an algorithm!

1	■ Take two slices of bread
2	■ Add butter or jam
3	■ Put the slices together
4	■ Eat your yummy sandwich!

6. Algorithms in Games ■

Games use algorithms to decide what happens next. Every move you make, every level you complete — the game follows steps to figure out what to do! Whether it is a board game or a computer game, algorithms are working behind the scenes to make it fun.

How a Board Game Uses an Algorithm

1. Roll the dice.
2. Move your piece the same number of spaces.
3. Follow the instruction on that space (go forward, go back, collect a card).
4. Pass the dice to the next player.

These ordered steps are an algorithm that makes the game work every time!

7. Computer Algorithms Help Us ■

Algorithms help computers do amazing things! They help find answers to your questions, show you fun videos, and play your favourite music. Every time you use a tablet or computer, algorithms are working behind the scenes!

- ■ Algorithms help computers find answers quickly.
- ■ They help show videos and play your favourite music.
- ■ Algorithms make apps and games work just right!
- ■ When you search for your favourite cartoon, an algorithm helps find it!

8. How Robots Use Algorithms ■

Robots are smart machines, but they need instructions! Every robot follows a set of steps — an algorithm — to do its job. Whether it is cleaning the floor, sorting toys, or moving objects, robots follow their algorithm carefully, step by step, every single time!

■ Robot Vacuum Cleaner Algorithm

1. Start in one corner of the room.
2. Move forward until you hit a wall or obstacle.
3. Turn and move in a new direction.
4. Repeat until the whole floor is clean.
5. Return to the charging station.

Without this algorithm, the robot would not know what to do!

9. Algorithms in Nature ■

Did you know nature has algorithms too? Nature follows patterns and repeated steps that work just like algorithms.

■ Bee collecting nectar

1. Fly to a flower.
2. Collect nectar.
3. Return to the hive.
4. Repeat!

■ Bird migration

1. Follow the sun's direction.
2. Fly south in winter.
3. Return north in spring.
4. Repeat each year!

■ Seed growing into a plant

1. Absorb water from soil.
2. Sprout a shoot upward.
3. Open leaves to sunlight.
4. Grow and flower!

■ **Did You Know?** Algorithms are not just for computers — they are all around us! From morning routines to cooking, games to nature — algorithms are everywhere!

10. Let's Make an Algorithm Together! ■

Can you think of the steps to pack your school bag? Let's make an algorithm together! Try writing the steps in order.

1	■ Put your books inside.
2	✏■ Add your pencil box.

3	■ Place your lunch box.
4	■ Don't forget your water bottle!

11. Classroom Activity ■■

Ask students to write their own algorithms for one of the following tasks. They should write at least 3–5 steps in the correct order.

- ■ Getting ready for bed
- ■■ Drawing a picture
- ■ Putting away your toys
- ■ Reading a book from the shelf

Write your algorithm here:

Step 1:

Step 2:

Step 3:

Step 4:

Step 5:

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12. Quiz Time! ■

Think carefully and answer these questions. Raise your hand when you know the answer!

Q1. What is an Algorithm?

Answer hint: An algorithm is a set of steps that helps us solve a problem or complete a task in the right order. Example: brushing your teeth step by step is an algorithm!

Q2. Name one algorithm you use every day.

Answer hint: Examples: morning routine, brushing teeth, making a sandwich, packing your school bag.

Q3. Why do robots need algorithms to work?

Answer hint: Robots need algorithms because they are machines — they cannot think on their own. Without a set of instructions (algorithm), a robot would not know what to do next!

Key Vocabulary

Word	Meaning
Algorithm	A set of steps to solve a problem or complete a task.
Sequence	Doing things one after another, in the right order.
Robot	A machine that follows instructions (algorithms) to do jobs.
Computer	An electronic device that uses algorithms to process information.
Pattern	Something that repeats in a regular, predictable way.

■ **Did You Know?** "Algorithms are like invisible helpers that make life easier!"

Keep exploring algorithms every day! ■

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